

A Compact 3–30-GHz 68.5-ps CMOS True-Time Delay for Wideband Phased Array Systems

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Abstract—This article presents a compact 4-bit switched-line true-time delay (TTD) circuit over a wide frequency range extending from 3 to 30 GHz using novel delay elements. The delay elements, namely, the cascading coupled all-pass network (CAPN) and noncoupled all-pass network (NCAPN), were employed in the proposed TTD circuit to improve the delay–bandwidth product (DBW) while maintaining its compact size and low delay variation (DV). For comparison, a theoretical analysis for understanding the group delay feature of the APN with various coupling coefficients is presented along with low-pass network (LPN). To verify the proposed structure, the proposed delays are applied to construct the 4-bit switched-line TTD by utilizing two single-pole double-through (SPDT) and three double-pole double-through (DPDT) switches in a 28-nm CMOS process. The circuit has a compact size of 1.7 mm × 0.2 mm, with a maximum delay of 68.5 ps and a minimum delay of 4.6 ps. The measured average insertion loss is 13.5 dB, and the in/out return loss is better than 10 dB across 3–30 GHz. The measured rms delay and gain errors are less than 2 ps and 3.2 dB, respectively, over the operating frequency range. To the best of our knowledge, the proposed TTD achieves the largest figure of merit (FoM) among the integrated TTD circuits.

Index Terms—All-pass network (APN), CMOS switch, delay variation (DV), phased arrays, true-time delay (TTD).

I. INTRODUCTION

PHASED-ARRAY techniques have become widespread in recent years, especially in the applications of both the incoming fifth-generation (5G) millimeter-wave (mm-wave) wireless communication technology [1], [2] and high-resolution radar systems [3]–[5]. Commonly used phased arrays employ phase shift by using reflection-type [6], [7], switched LC-type [8]–[10], and vector modulator-type configurations [11]–[13] to approximate the required time delay in a narrow bandwidth. It is well known that conventional phased arrays suffer from the beam-squinting phenomenon. To address

this issue, the true-time delay (TTD) circuit is therefore a critical block that should be applied in wideband phased arrays. Although various time-delay techniques with nonsilicon solutions have been proposed in the literature [14]–[19], many efforts have been made to implement TTD within silicon-based technologies for fully integrated and low-cost systems. However, the implementation of TTD in silicon-based technologies has several limitations, such as its small time delay, bandwidth, and large chip size. Recently, several works involving integrated TTDs have been proposed to overcome these shortcomings.

Many types of integrated TTDs have been developed as a varactor-loaded and a switched-line configuration using low-pass networks (LPNs) [21]–[25], which are the commonly used techniques of TTD structures. However, due to the existing cutoff frequency of LPNs, a low delay per unit cell should be selected and cascaded to be a form of a multistage structure in order to achieve a large total variable delay, maintaining a low delay variation (DV) over the wide bandwidth, which leads to a large footprint and high insertion loss. In other words, it is difficult to realize a large delay–bandwidth product (DBW) and low DV with a compact size in the field of integrated circuits using conventional LPN methods.

To overcome the drawback of the LPN, the use of all-pass networks (APNs) is considered an emerging approach for implementing the TTD blocks. The APN circuits can be realized using both active and passive delay topologies. The active APN topology has been realized by incorporating transistor elements (G_m) with passive elements (R and C) as the first- and second-order transfer functions [26], [27]. Up until now, this topology has been operated in low-frequency bands, even with a small chip area because of the significant parasitic effects of transistor elements at mm-wave frequencies. The passive topology in [28] comprises only magnetically coupled inductors and capacitors, which can be included in the passive and active topologies, such as switch-embedding type [29] and trombone-type [30] delay circuits. Although the APN topology has a larger delay per unit cell than that of the LPN topology, it also has a limited DBW, especially in cases that require large constant delays.

As a result, each reported TTD has several disadvantages. The active APN-based TTD can be used only at low-frequency bands with power consumption and low linearity, while it occupies a small area. The prevalent configuration, which is implemented by the LPN delay, should be used only by cascading multistage delay cells because of the existing cutoff frequency. In addition, the switch-included passive APN

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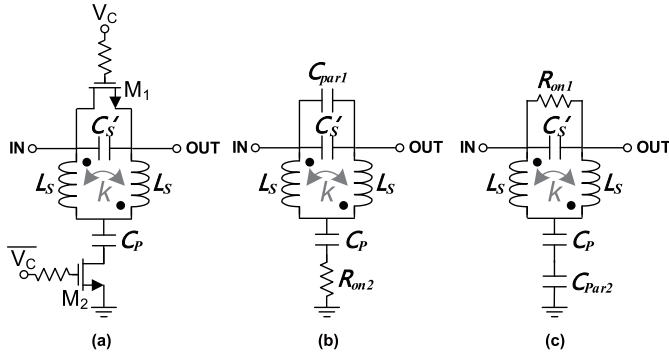


Fig. 1. (a) Conventional switch-embedded TTD circuit, (b) its delay state, and (c) its bypass state [29].

configuration [29] also has a limited DBW and is vulnerable to the finite ON-resistance of the switch when it is required to have a large nominal delay in the high-frequency band. Therefore, there is a need for a compact TTD with a large DBW that has no power consumption and high linearity.

In this article, we present a compact 4-bit switched-line TTD operating from 3 to 30 GHz using the proposed novel delay elements. To synthesize the novel delay elements, a delay compensation between the cascaded coupled APN (CAPN) and noncoupled APN (NCAPN) is adopted for a large DBW per size with low DV. This article is organized as follows. Section II discusses the delay characteristics of APN and compares it with a unit LPN structure. Furthermore, using this analysis, the proposed novel delay elements are constructed. Section III presents the measured and simulated results of the stand-alone proposed delays and single-pole double-through (SPDT)/double-pole double-through (DPDT) switches. In Section IV, the fabrication of the proposed 4-bit TTD is discussed, and the design is validated using the measured and simulated results.

II. COMPARISON OF TIME-DELAY CIRCUITS

A. Conventional APN-Based Switch-Embedded TTD Circuit

A typical topology for a 1-bit TTD circuit is the switched-line TTD using a pair of SPDT switches at both the input and output sides to provide the digitally controlled delay. To lower the insertion loss from finite ON-resistances (R_{ON}) of the series-connected switches and to reduce the circuit size, a switch-embedded topology is also commonly used to implement the 1-bit TTD circuit. Fig. 1(a) shows the configuration of the 1-bit APN-based TTD circuit [29]. This circuit has two switches that are controlled by a control voltage signal V_C and is operated in the delay state and bypass state, as shown in Fig. 1(b) and (c), respectively. In the delay state, M_1 is OFF and M_2 is ON. Neglecting switch parasitics, this state has the form of the classical passive second-order APN, which could offer a nominal delay. In the bypass state where M_1 is ON and M_2 is OFF, the circuit offers zero delay.

Fig. 2(a) and (b) shows the equivalent circuits of the conventional switch-embedded TTD in the delay state and bypass state, respectively. Considering switch parasitics, the equivalent circuits of both states have R_{ON} (finite ON-resistance of

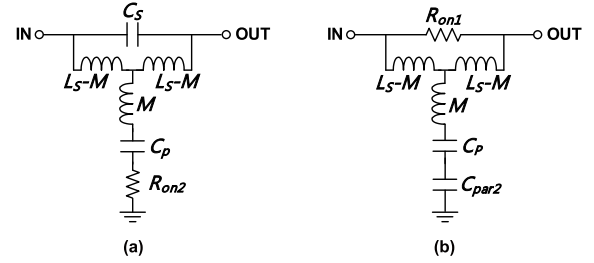


Fig. 2. Equivalent circuits of the conventional switch-embedded TTD in (a) delay state and (b) bypass state.

the used switch) and C_{par} (parasitic OFF-capacitance of the used switch). The mutual inductance M can be expressed as

$$M = kL_S \quad (1)$$

where k is defined as the coupling coefficient between two inductors (L_S) of the APN. Because the equivalent circuits of the CAPN in both states shown in Fig. 2(a) and (b) are symmetrical, even- and odd-mode analyses [28] can be performed to obtain the S-parameters as follows:

$$S_{11} = \frac{1}{2}(\Gamma_{\text{even}} + \Gamma_{\text{odd}}) = \frac{Z_{\text{even}}Z_{\text{odd}} - Z_o^2}{(Z_{\text{even}} + Z_o)(Z_{\text{odd}} + Z_o)} \quad (2a)$$

$$S_{21} = \frac{1}{2}(\Gamma_{\text{even}} - \Gamma_{\text{odd}}) = \frac{Z_{\text{even}}Z_o - Z_{\text{odd}}Z_o}{(Z_{\text{even}} + Z_o)(Z_{\text{odd}} + Z_o)} \quad (2b)$$

where Z_o is the characteristic impedance and $Y_{\text{odd}} = 1/Z_{\text{odd}}$. The even- and odd-mode impedances in the delay state shown in Fig. 2(a) are expressed as

$$Z_{\text{even}} = j\omega(L_S + M) + \frac{2}{j\omega C_P} + 2R_{ON2} \quad (3a)$$

$$Y_{\text{odd}} = j\omega(2C_S) + \frac{1}{j\omega(L_S - M)}. \quad (3b)$$

In the delay state, the specific condition of $Z_{\text{even}}/Y_{\text{odd}} = Z_o^2$ should be satisfied in order to fulfill $S_{11} = 0$ at all frequencies. As a result, the following relation can be obtained as:

$$\frac{1+k}{1-k} = \frac{4C_S}{C_P}. \quad (4)$$

By applying the relation, $Z_{\text{even}}/Y_{\text{odd}} = Z_o^2$, and (3a) into (2b), S_{21} can be expressed in the following equation:

$$S_{21} = \frac{(j\omega)^2 - \frac{Z_o}{L_S(1+k)}(j\omega) + \frac{2}{L_S(1+k)C_P}}{(j\omega)^2 + \frac{Z_o}{L_S(1+k)}(j\omega) + \frac{2}{L_S(1+k)C_P}}. \quad (5)$$

It can be seen that S_{21} is constant and has unity gain when all switch parasitics are neglected. At low frequencies, the nominal group delay is defined by using the general formula defined in (5). As in the delay state, even- and odd-mode analysis is conducted again in the bypass state, which is shown in Fig. 2(b)

$$Z_{\text{even}} = j\omega(L_S + M) + \frac{2}{j\omega(C_P || C_{par2})} \quad (6a)$$

$$Y_{\text{odd}} = \frac{2}{R_{ON1}} + \frac{1}{j\omega(L_S - M)}. \quad (6b)$$

For simplicity, R_{ON} is assumed to be very low, which leads to $Z_{even} = 0$, and S_{21} and S_{11} are only affected by Z_{even} . By using (2b), S_{21} in the bypass state can be expressed in the following equation:

$$S_{21} = \frac{(j\omega)^2 + \frac{2}{L_s(1+k)C_{p'}}}{(j\omega)^2 + \frac{Z_o}{L_s(1+k)}(j\omega) + \frac{2}{L_s(1+k)C_{p'}}} \quad (7)$$

where $C_{p'} = C_p || C_{par2}$ and the nominal group delay is defined by using the general formula defined in (7) at low frequencies. If the network in the bypass state has to fulfill $S_{11} = 0$ at all frequencies, the value of C_{par2} has to be zero or very small. However, this requirement leads to a large R_{ON2} , i.e., switch with a small width, which further deteriorates S_{11} and the DV in the delay state [29] if the conventional switch-embedded TTD should be designed with a larger delay. In other words, the conventional 1-bit switch-embedded TTD can offer a limited nominal delay with a high DV versus frequency, i.e., limited DBW due to the parasitic resistance and capacitance. A multistage switch-embedded configuration or the switched-line TTD configuration should be used to realize a large switching delay. In this article, the switched-line configuration is chosen to implement the TTD block with high DV versus frequency performance. In Section II-A, various delay elements will be compared in terms of DV versus frequency and the DV compensated method is employed to propose the novel delay elements with high DV versus frequency performance.

B. Comparison of Various Delay Elements

In this section, the delay characteristics of the APN is carefully analyzed and compared with that of a conventional unit LC-network, which is denoted as LPN. Fig. 3 shows the schematics of the unit stage CAPN, NCAPN, and LPN, where NCAPN is separately presented using a series-L configuration, which is a special case of CAPN ($k = 0$) [28]. The absolute group delays of three structures are defined as

$$\tau_{CAPN} = \frac{2}{\omega_{T1}Q} \times \frac{1 + (\omega/\omega_{T1})^2}{[1 - (\omega/\omega_{T1})^2]^2 + [\omega/(\omega_{T1}Q)]^2} \quad (8a)$$

$$\tau_{NCAPN} = \frac{2}{\omega_{T2}} \times \frac{1 + (\omega/\omega_{T2})^2}{[1 - (\omega/\omega_{T2})^2]^2 + (\omega/\omega_{T2})^2} \quad (8b)$$

$$\tau_{LPN} = -\frac{1}{\omega} \tan^{-1}(\omega/\omega_C) \times \frac{2 - (\omega/\omega_C)^2}{1 - 2(\omega/\omega_C)^2} \quad (8c)$$

$$\omega_{T1} = \frac{2}{\sqrt{2L_{T1}C_{T1}}} = \frac{2}{QZ_oC_p} \quad (9a)$$

$$\omega_{T2} = \omega_C = \frac{1}{\sqrt{LC}} \quad (9b)$$

where ω_{T1} and ω_{T2} are the transition frequencies of CAPN and NCAPN, respectively, and ω_C is the cutoff frequency of LPN. In (9a), the relations, $L_{T1} = L_s + M$ and $C_{T1} = C_p$, were used to define the transition frequency of CAPN (ω_{T1}). It can be observed that when the operating frequency is much less than the transition and cutoff frequencies, (8a)–(8c), in this case, are simplified to $\tau_{CAPN} \approx C_p Z_o$, and $\tau_{NCAPN} \approx \tau_{LPN} \approx 2\sqrt{LC}$, where the delay elements can provide constant nominal delays at low frequencies.

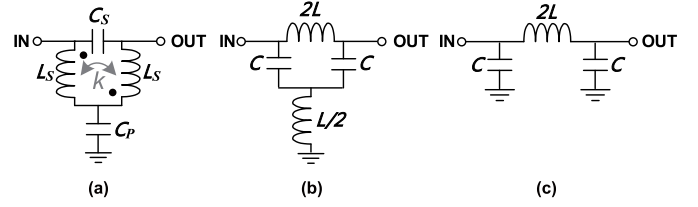


Fig. 3. Schematics of (a) single-stage CAPN, (b) NCAPN, and (c) LPN.

As shown in Section II-A, due to the limited DBW of those three delay networks, the DV from the nominal value occurs as the operating frequency increases. For comparison, Fig. 4(a) shows the simulated group delay characteristic of the unit delay networks, CAPN, NCAPN, and LPN, where the nominal delays are set to be 6 ps. As expected, DV becomes more severe as the operating frequency gets closer to the cutoff and transition frequencies. Interestingly, the delay response of CAPN can be varied for various k values, and the DV can be minimized when $k = 0.5$ in this state. However, when the nominal delay is chosen to be larger than 12 ps, the DV still exists, as shown in Fig. 4(b). The reason is that the phase shift of the unit CAPN can approach but not exceed 360° as the frequency increases infinitely. Therefore, the delay response should slide down at a specific point as the frequency increases. This feature requires that the conventional CAPN delay cell should also be used in a form of multistage network in order to overcome the DV. Therefore, the DV compensation technique should be adopted to circumvent a large footprint by cascading the CAPN for $k = 0.5$, which has a gradually declining delay, and NCAPN, with an expansive delay, as the frequency increases, instead of using the multistage technique.

III. DESIGN AND IMPLEMENTATION

In this section, the delay compensation technique is employed to implement novel delay elements that have an improved DBW with compact size. The proposed novel delay elements are also applied to realize a 4-bit switched-line TTD circuit.

A. Delay Elements Based on Delay Compensation Technique

Before cascading CAPN and NCAPN delay cells, various delay elements should be compared using the configurations shown in Fig. 3. To estimate how the delay elements have low DVs over the frequency range, the normalized DV (NDV) defined in [31] is utilized

$$NDV(\%) = \frac{\max(\tau_\omega) - \min(\tau_\omega)}{\tau_o} \times 100. \quad (10)$$

With the definition, the proposed delay elements are implemented by cascading an appropriate number of CAPN and NCAPN cells to have the lowest NDV among other delay elements over 3–30 GHz. Since each CAPN and NCAPN cell is not appropriate for the 12- and 24-ps delay element as shown in Fig. 4, the multistage or DV compensation technique should be employed. Fig. 5 shows the schematics of various 12-ps delay elements and 24-ps delay elements. The proposed 12-ps delay (CAPN + NCAPN) is composed of an 8.86-ps CAPN cell and a 3-ps NCAPN cell, and the other proposed

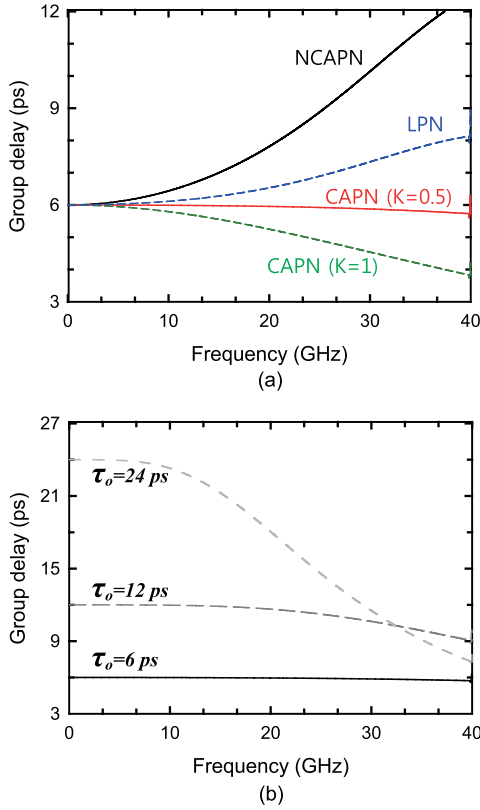


Fig. 4. Group delay characteristic for (a) single-stage circuits in Fig. 3 and (b) different nominal delays of CAPN with $k = 0.5$.

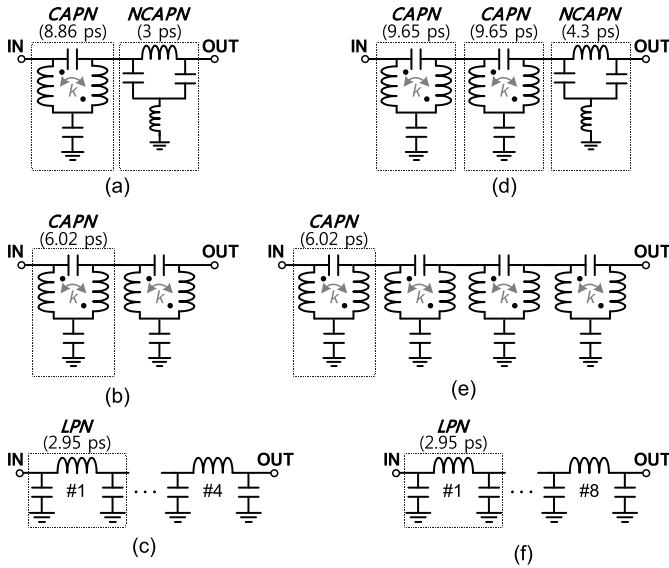


Fig. 5. Schematics for 12-ps delay elements cascading (a) CAPN and NCAPN, (b) two CAPNs, and (c) four LPNs, and 24-ps delay elements cascading (d) two CAPNs and NCAPN, (e) four CAPNs, and (f) eight LPNs.

24-ps delay consists of two 9.65-ps CAPN cells and a 4.3-ps NCAPN cell. For the 12-ps delay elements, delay elements cascading two CAPNs with $k = 0.5$ (2*CAPNs), where each stage has 6.02 ps, and four LPNs (4*LPNs), where each stage has 2.95 ps, are compared with the proposed delay element. As shown in Fig. 6(a), the proposed 12-ps delay element has the lowest NDV of 1.1%, which is lower than 3.4% and 7.8% of 2*CAPNs and 4*LPNs, respectively. In addition,

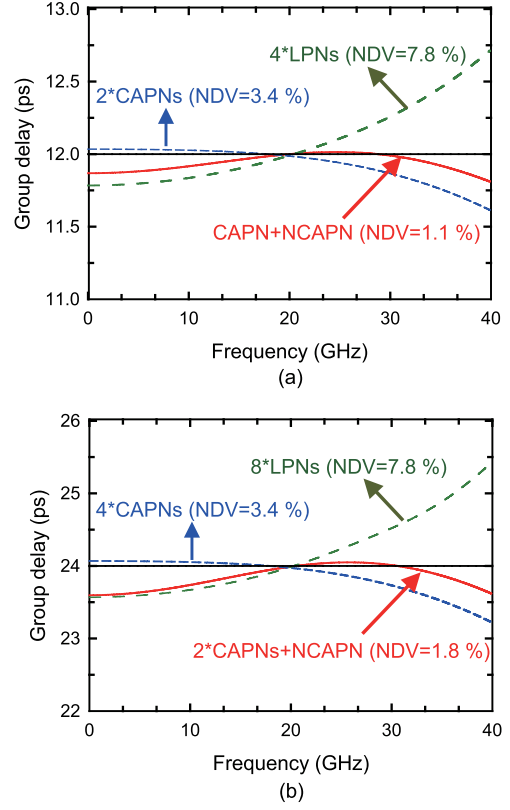


Fig. 6. Simulated delay responses of the various cascaded delay elements shown in Fig. 5 for (a) 12-ps delay and (b) 24-ps delay.

for the 24-ps delay elements, delay elements cascading four CAPNs with $k = 0.5$ (4*CAPNs) and eight LPNs (8*LPNs) are compared with the proposed delay element. As shown in Fig. 6(b), the NDV of 2*CAPNs + NCAPN is 1.8%, which is the lowest NDV value compared to 4*CAPNs and 8*LPNs. Furthermore, the proposed delay element shown in Fig. 5(a) and (d) is composed of only two and three sections, respectively. Because each CAPN with high mutual coupling coefficient ($k = 0.5$), NCAPN, and LPN section has almost the same physical size, the less sections of delay element give the size reduction. Therefore, the proposed delay elements can accomplish both size reduction and group delay flatness for a given bandwidth (3–30 GHz) and can be utilized to implement a compact wideband TTD chain.

B. Design of the Proposed 4-bit Switched-Line TTD

Based on the proposed delay elements, the design of a compact 4-bit switched-line TTD circuit is presented in this section. All of the delay elements used in the TTD circuit are shown in Fig. 7(a). Considering the delay of the reference line of the switched-line TTD circuit, the 24-ps delay element illustrated in Section III-A is shortened to a 23-ps delay element. A 46-ps delay element is composed of two cascading 23-ps delay elements, and for brevity, this delay element is therefore not shown in this figure. The design parameters of the delay elements in Fig. 7(a) are listed in Table I, and the simulated Q factor of the each inductor is presented in Fig. 7(b). The Q -factor of each inductor is calculated from the electromagnetic (EM) simulation. All the capacitors in the

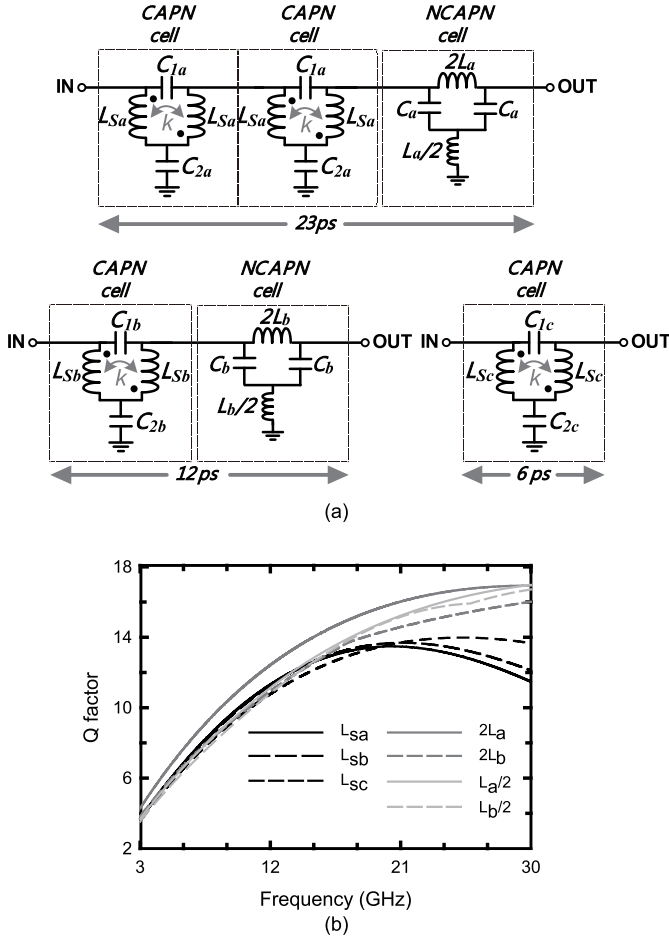


Fig. 7. (a) Schematic of the proposed delay elements and (b) simulated Q of the used inductors. For brevity, the 46-ps delay is not shown.

TABLE I
DESIGN PARAMETERS OF DELAY ELEMENTS IN FIG. 7

L_{sa}	C_{1a}	C_{2a}	L_a	C_a	L_{sb}	C_{1b}	C_{2b}	L_b	C_b	L_{sc}	C_{1c}	C_{2c}
160.8	16.1	193	107.5	43	147.8	14.8	177.4	75	30	100	10	120
(pH)	(fF)	(fF)	(pH)	(fF)	(pH)	(fF)	(fF)	(pH)	(fF)	(pH)	(fF)	(fF)

delay elements were implemented using the PDK model of the 28-nm process (Q -factor > 20).

The proposed 4-bit switched-line TTD consists of two SPDT switches, three DPDT switches, and delay elements that contain the novel delay elements, as shown in Fig. 7. The single DPDT switch is used to reduce the insertion loss from switches compared with that from a pair of SPDT switches by reducing the number of series transistors in the signal path. Fig. 8(a) shows the circuit schematic of the SPDT switch using a series-shunt configuration to improve isolation, where the total width of the series and shunt transistors is chosen to 80 and 45 μm , respectively. Furthermore, the DPDT switch is configured with four series and cross-connected transistors having a total width of 40 μm , as shown in Fig. 8(b). All the switches are realized using the body floating technique to improve the insertion loss. Fig. 9 shows the configuration of the proposed 4-bit switched-line TTD. The CAPN and NCAPN cells are implemented using the layout configuration shown in Fig. 10. Because the used delay elements occupy a

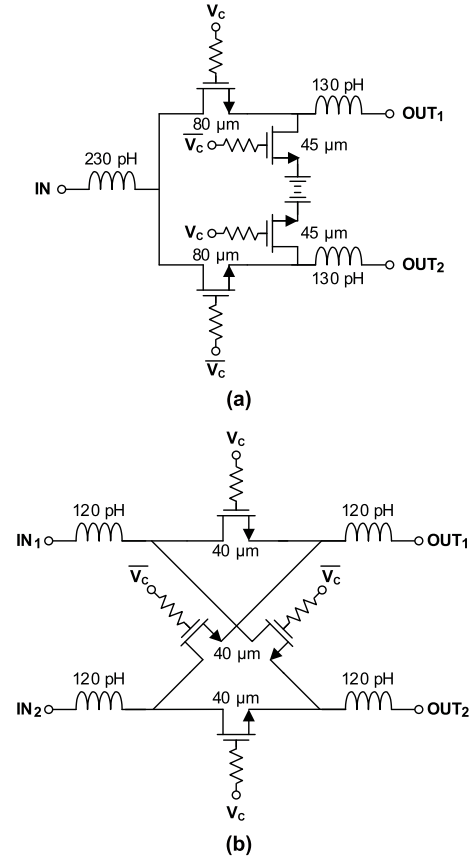


Fig. 8. Schematic of (a) SPDT and (b) DPDT switch.

small chip area, and three pairs of SPDT switches are replaced with only three DPDT switches, the 4-bit switched-line TTD chain can be realized in a compact footprint. The TTD circuit can shift the total delay of 68.5 ps with a resolution of 4.6 ps.

IV. EXPERIMENTAL VERIFICATION

The proposed 4-bit TTD chain and stand-alone blocks, such as the delay element and switch blocks, were fabricated using a 28-nm CMOS technology. All inductors and interconnecting lines were simulated using Sonnet considering all parasitics. Fig. 11 shows the chip microphotograph of the stand-alone delay element blocks of 46, 23, 12, and 6 ps, which occupy core areas of $470 \times 160 \mu\text{m}^2$, $235 \times 160 \mu\text{m}^2$, $155 \times 146 \mu\text{m}^2$, and $73 \times 103 \mu\text{m}^2$, respectively, excluding pads. The S-parameter measurements were performed using an Agilent E8364A network analyzer by the SOLT calibration to the GSG 100-pitch probe tips.

Fig. 12(a) shows the measured and simulated S-parameter and group delay results of each delay block. The difference between the measured and simulated insertion loss is getting more distinct as the delay block becomes large. The measured delay blocks have a wideband matching characteristic with $S_{11} < -20$ dB over 3–30 GHz. The measured insertion loss values of 46-, 23-, 12-, and 6-ps delay blocks are 5, 2.5, 1.1, and 0.45 dB, respectively, at 28 GHz. The measured and simulated group delay curves corresponding to each delay block are shown in Fig. 12(b). For the group delay measurement, the number of sampling points and IF bandwidth

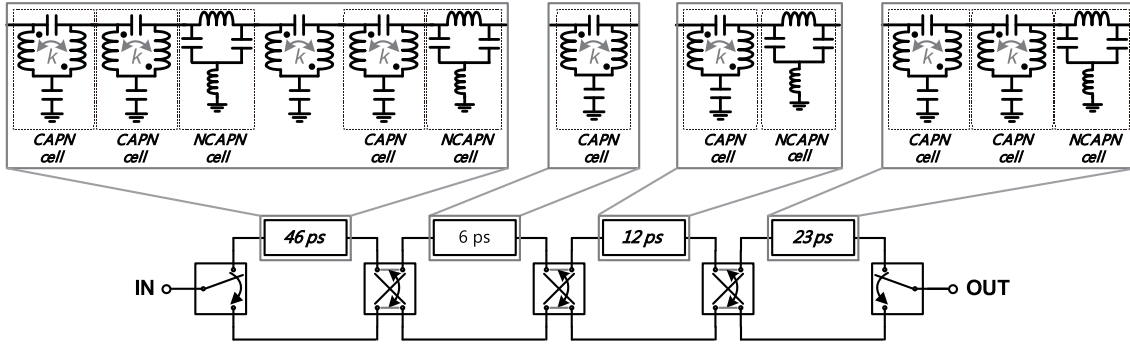


Fig. 9. Configuration of the proposed 4-bit switched-line TTD chain.

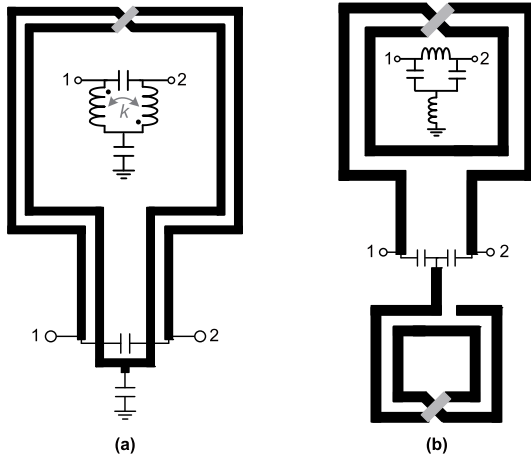


Fig. 10. Layout configurations of inductor components for (a) CAPN cell and (b) NCAPN cell.

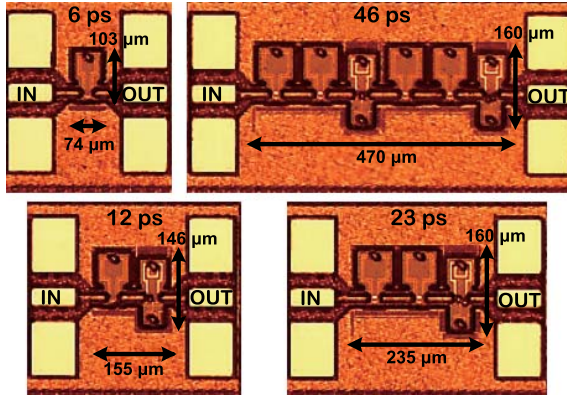


Fig. 11. Chip microphotographs of stand-alone delay blocks.

resolution is chosen to be 1601 and 10 kHz, respectively, to a lower degradation from the noise level. The group delay data were acquired by a derivative of the measured phase responses over 3–30 GHz.

Fig. 13 shows the chip microphotograph of the test SPDT and DPDT switches, and the proposed 4-bit TTD. The core areas of the SPDT and DPDT switches are $175 \times 134 \mu\text{m}^2$ and $129 \times 134 \mu\text{m}^2$, respectively. With the Out_1 port loaded with a $50\text{-}\Omega$ coaxial termination, the Out_2 port is measured as the signal path. The measured and simulated insertion loss and the return loss of the SPDT switch are shown in Fig. 14. The insertion loss is 1.7 dB with S_{11} and $S_{22} < -17$ dB, and the isolation is 19 dB at 28 GHz when the

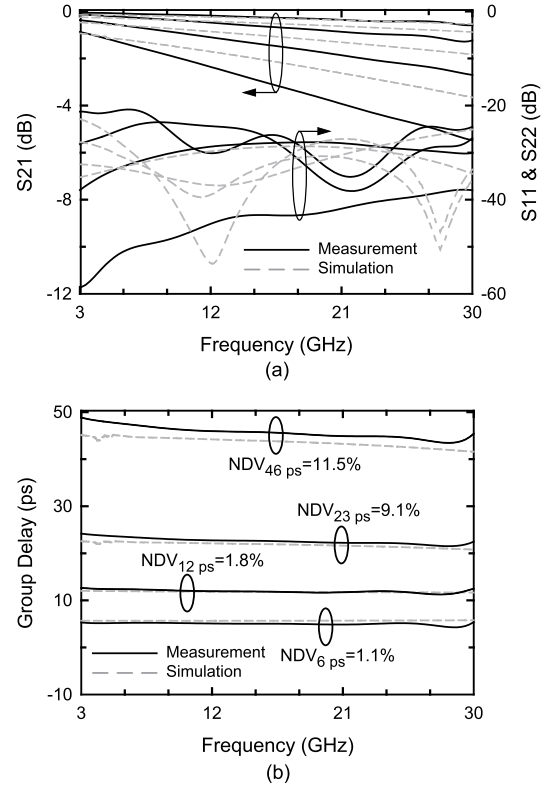


Fig. 12. Measured and simulated (a) S-parameters and (b) group delay responses of each delay block in Fig. 11.

switch operates in ON- and OFF-states, respectively. Due to the additional resistance and inductance of the routing lines and ground vias in the shunt switch, the measured isolation and input–output return losses were degraded compared with the simulation results. As opposed to the case with the SPDT switch, the DPDT switch was measured twice to determine the transmission- and cross-path response. The transmission path is chosen between the In_1 port and Out_1 port, and the cross path is chosen from the In_1 port to Out_2 port, with all remaining ports being loaded with $50\text{-}\Omega$ termination, as with the SPDT switch. The measured and simulated results of the DPDT switch are shown in Fig. 15. The insertion loss is 2.4 dB with S_{11} and $S_{22} < -15$ dB, and the isolation is 18 dB at 28 GHz when the switch operates in ON- and OFF-states, respectively.

Using the delay blocks, including the novel delay elements, and SPDT and DPDT switches, the proposed 4-bit

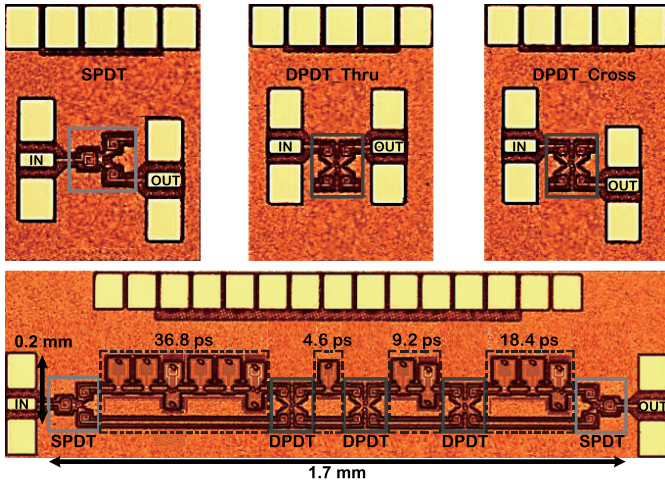


Fig. 13. Chip microphotographs of stand-alone SPDT, DPDT switch, and 4-bit switched-line TTD chain.

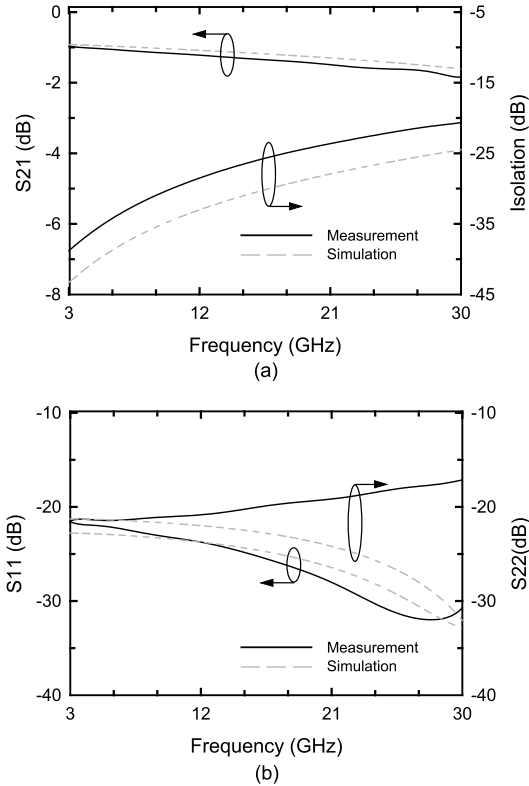


Fig. 14. Measured and simulated S-parameters of the SPDT switch. (a) Insertion loss and isolation. (b) Input and output return loss.

switched-line TTD chain is fabricated, as shown in Fig. 13. The TTD chain is composed of four delay blocks, two SPDT switches, and three DPDT switches with a circuit size of $1.7 \text{ mm} \times 0.2 \text{ mm}$. The measured S-parameters of the proposed TTD chain are shown in Fig. 16, which can operate over the wideband frequency range from 3 to 30 GHz. Because of the difference between the measured and simulated insertion losses of the delay blocks as shown in Fig. 12(a), the loss variation inevitably rises. The average insertion loss of 16 different states ranges from 7.1 to 16.7 dB in the operation frequency and is about 14 dB at 28 GHz. In addition, the measured results of both S_{11} and S_{22} are lower than

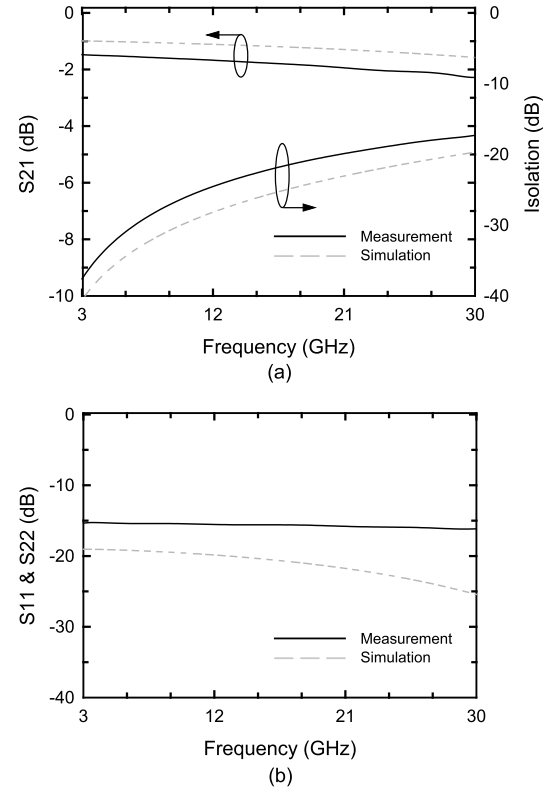


Fig. 15. Measured and simulated S-parameters of the DPDT switch. (a) Insertion loss and isolation. (b) Input and output return loss.

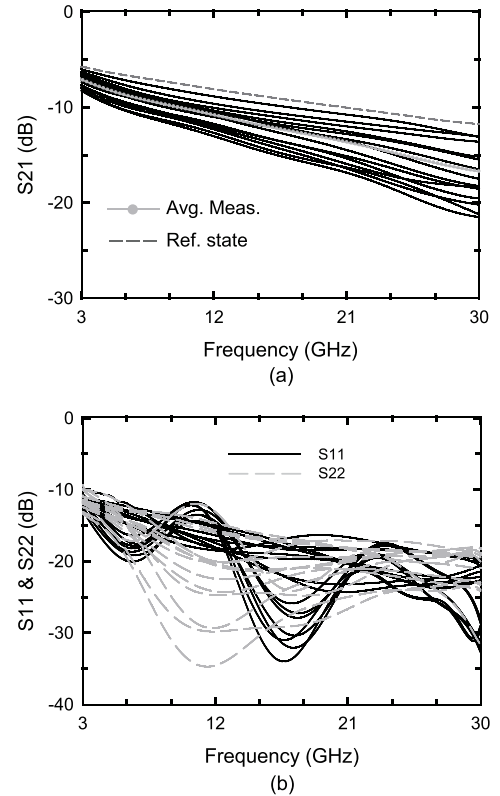


Fig. 16. Measured (a) insertion loss and (b) input and output return loss of 16 different states.

−10 dB within the operation frequency. Because the measured insertion loss of the DPDT switch is 0.65 dB higher than the simulated one, the insertion loss of the TTD with three DPDT

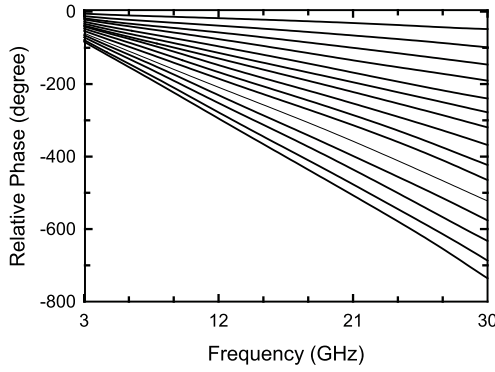


Fig. 17. Measured relative insertion phase responses.

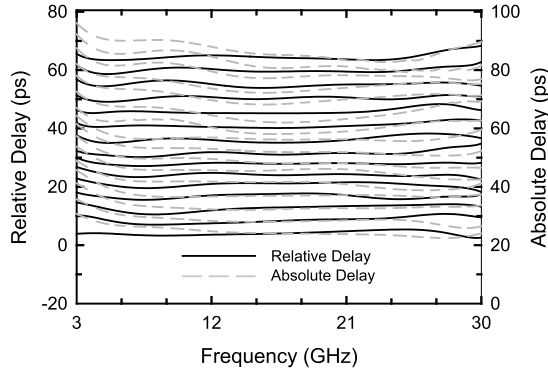


Fig. 18. Measured group delay curves.

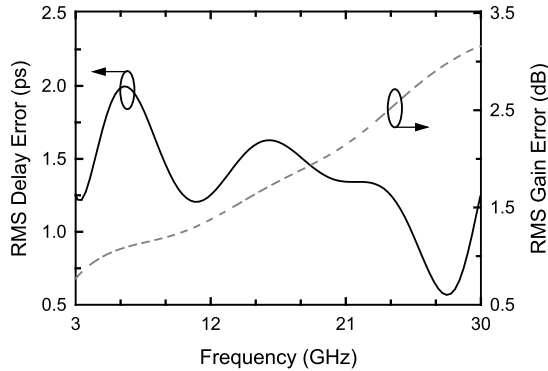
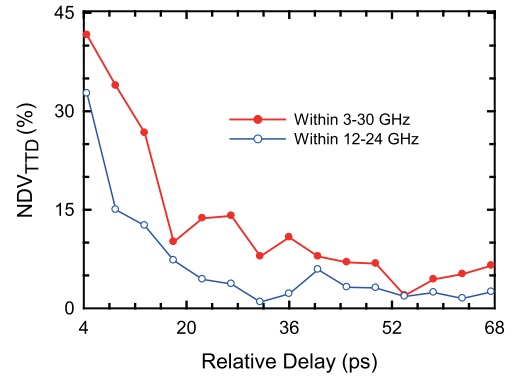
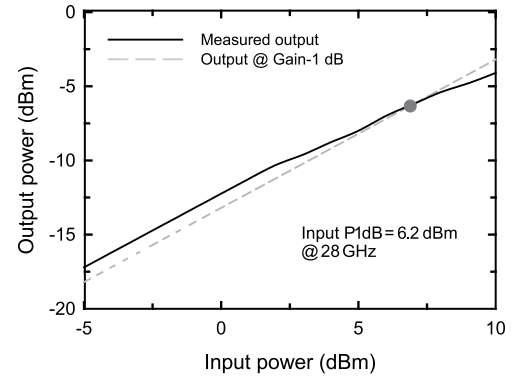


Fig. 19. Calculated rms delay and gain error.

switches is higher than the expected one by almost 2 dB. The insertion loss of the TTD can be reduced by optimizing the nMOS switch size and the parasitic resistances in the DPDT switches. The optimal switch size is related to a tradeoff between the insertion loss (R_{ON}) and the isolation (C_{OFF}), as explained in [35]. Therefore, the series-shunt configuration needs to be applied in order to improve the insertion loss of the DPDT switch.

Fig. 17 shows the measured relative phase performance of the TTD chain. It can be observed that the relative phase shift at 28 GHz reaches 675° , that is, 720° with a resolution of 45° . This feature makes the proposed TTD chain more attractive than the commonly used phase shifter when a large phased array is required. The operating phase shift range of 720° is demonstrated for the first time in an mm-wave using a standard CMOS process. The absolute and relative group delays are measured, as shown in Fig. 18. The maximum relative delay is

Fig. 20. Calculated NDV_{TTD} of the proposed TTD chain.Fig. 21. Measured IP_{1dB} of the TTD chain.

68.5 ps with a 4-bit resolution of 4.6 ps across 3–30 GHz. This performance verifies that the proposed chain can be employed in the wideband phased-array system without suffering from the beam-squinting phenomena. The rms delay and gain errors are calculated using the measured results, as shown in Fig. 19. The rms delay errors for the TTD are calculated using [25] as follows:

$$\tau_{\Delta, \text{rms}} = \sqrt{\frac{1}{N-1} \times \sum_{i=2}^N |\tau_{\Delta i}|^2} \quad (\text{ps}) \quad (11)$$

where $N = 16$ and $\tau_{\Delta i}$ means the i th output delay error from the ideal delay value. The rms delay and gain errors are, respectively, less than 2 ps and 3.2 dB from 3 to 30 GHz. The proposed TTD obtains an in-band NDV_{TTD} of $<15\%$ except for three delay states, i.e., 4.6-, 9.2-, and 13.8-ps delay states from 3 to 30 GHz, as shown in Fig. 20. When the operating frequency range is scaled down to 12–24 GHz, which still covers an octave frequency range, the in-band NDV_{TTD} of $<7.4\%$ can be achieved, with the exception of the three delay states (4.6, 9.2, and 13.8 ps).

The input 1-dB gain compression point (IP_{1dB}) is measured, as shown in Fig. 21. The measured IP_{1dB} is around 6.2 dBm at 28 GHz. This value is limited by the junction diode of the CMOS SPDT and DPDT switches. Tables II and III compare the proposed TTD chain with other previously reported silicon and nonsilicon-based TTD chains. In order to show the properties of the proposed TTD effectively, DBW that is defined

TABLE II
PERFORMANCE COMPARISON WITH NONSILICON TECHNOLOGIES

Ref.	Tech.	Maximum delay (ps)	Freq. (GHz)	Nominal [†] Loss (dB)	DBW	Minimum delay (ps)	RMS delay error (ps)	Size (mm ²)	Power (mW)	FoM (1/mm ²)
This work	28 nm CMOS	68.5	3-30	13.5 (7.1-16.7)	1.85	4.9	< 2.0	0.34	Passive	116
[15]	GaAs MEMS	25	0-40	1.35 (0.5-2.2)*	1.0	3.57	N/A	9.1	Passive	80.5
[17]	MMIC MEMS	21.9*	15-40	5.6 (3.0-8.2)	0.55	3.12*	N/A	63.7	Passive	1.35
[18]	GaAs pHEMT	145	2-20	14.3 (6.6-22)*	2.61	2.3	< 2.4	5.6	< 30	17.3
[20]	MMIC MEMS	159	13-18	2.25 (1.5-3)	0.79	4.06	N/A	13	Passive	30.6

[†]Mean value of the average insertion loss over the operating frequency range. *Estimated average insertion loss. *Calculated group delay at 40 GHz.

TABLE III
PERFORMANCE COMPARISON WITH SILICON TECHNOLOGIES

Ref.	Tech.	Maximum delay (ps)	Freq. (GHz)	Nominal [†] Loss (dB)	DBW	Minimum delay (ps)	RMS delay error (ps)	Size (mm ²)	Power (mW)	FoM (1/mm ²)
This work	28 nm CMOS	68.5	3-30	13.5 (7.1-16.7)	1.85	4.9	< 2.0	0.34	Passive	116
[23]	0.18 μ m CMOS	106	5-20	22.5 (12.5-32.5)*	1.59	3.3	< 1.0	0.88	Passive	1.01
[29]	0.18 μ m CMOS	125	8-18	19.2 (15.2-23.3)	1.25	3.9	< 3.3	2.0	Passive	2.9
[30]	0.13 μ m CMOS	400	1-20	27 (16-38)*	7.6	Cont.	N/A	4.0	2.6-6	0.3
[32]	0.13 μ m CMOS	200	8-15	<40	1.4	1.56	N/A	1.21	Passive	0.1
[33]	0.13 μ m CMOS	42	15-40	14 (12-16)	1.05	3	N/A	0.99	8.6-24.6	26.6
[34]	0.25 μ m SiGe	32.8	10-50	17 (13-21)*	1.31	Cont.	N/A	0.22	Passive	47.4
[35]	65 nm CMOS	35.2	20-30	12 (11-13)	0.35	0.56	< 0.7	0.18	Passive	98

[†]Mean value of the average insertion loss over the operating frequency range. *Estimated average insertion loss.

as the product of the maximum delay and bandwidth, and a figure of merit (FoM) of the proposed TTD is also utilized in the comparison tables. The FoM used in Tables II and III is applied for all the references, which is defined as

$$\text{FoM} = \frac{\Delta\text{GD}_{\text{max}} (\text{ps}) \times \text{BW} (\text{GHz})}{\text{Loss}_{\text{max}} \times \text{Size} (\text{mm}^2)} \times 1000 \quad (12)$$

where $\Delta\text{GD}_{\text{max}}$ is the maximum (relative) delay and Loss_{max} is the maximum average insertion loss over the bandwidth shown in Tables II and III, which is expressed in real numbers. The FoM consists of the DBW ($\text{BW} \times \Delta\text{GD}_{\text{max}}$), size, and Loss_{max} terms because many efforts have been focused to improve both of them in the previous works. The minimum delay term is not included for two main reasons. The minimum delay of an analog TTD cannot be defined unless this circuitry contains a digital-to-analog converter. In addition, the least significant bits, which are below 4.6 ps, do not degrade the FoM significantly. It can be seen that the proposed design features the state-of-the-art FoM among not only silicon but also nonsilicon-based integrated TTD circuits.

V. CONCLUSION

A compact 4-bit wideband switched-line TTD operating over 3–30 GHz was presented using novel delay elements. The proposed delay elements are composed of the cascaded CAPN and NCAPN cells to compensate for the DV. The proposed TTD chain was demonstrated using the 28-nm CMOS technology. The TTD chain provides 4-bit delay control with an MSB value of 36.8 ps and an LSB value of 4.6 ps with an rms delay and gain errors less than 2 ps and 3.2 dB, respectively. Compared with other reported integrated circuits, the proposed TTD has the highest FoM to-date. The results show that the proposed TTD is an attractive option for a wideband phased-array antenna system.

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